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Original Article

New-Onset Atrial Arrhythmias Are Independently Associated With In-Hospital Mortality in Veno-Venous Extracorporeal Membrane Oxygenation

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Objective: To explore if atrial arrhythmias are associated with in-hospital mortality in veno-venous extracorporeal membrane oxygenation (VV-ECMO) patients.
Design: Retrospective observational cohort study.
Setting: Quaternary care academic medical center.
Participants: Patients with respiratory failure requiring VV-ECMO for >24 hours between January 1, 2016, and January 1, 2019.
Interventions: None, observational study.
Measurements and Main Results: Two hundred nineteen VV-ECMO patients were included. Patients were stratified by absence or presence of clinically significant atrial arrhythmias during the VV-ECMO run. Atrial arrhythmias were defined as either atrial fibrillation or atrial flutter that occurred during VV-ECMO and required pharmacologic or electrical intervention. The primary outcome was in-hospital mortality. Secondary outcomes included a composite of thrombotic events, which included ischemic stroke and on-pump arterial thrombosis. Other objectives of this analysis included characterization of atrial arrhythmia incidence, risk factors, and management. A total of 67 patients (30.5%) experienced new-onset atrial arrhythmias post−ECMO cannulation. Age, male sex, and norepinephrine use were independently associated with atrial arrhythmia development. In-hospital mortality was significantly higher in the atrial arrhythmia group (38.8% *v* 19.1%; p = 0.003). In the multivariate logistic regression analysis, atrial arrhythmias during VV-ECMO were independently associated with increased odds of in-hospital mortality (odds ratio, 2.21; 95% confidence interval, 1.08-4.55; p = 0.03), after controlling for Respiratory Extracorporeal Membrane Oxygenation Survival Prediction score, acute renal failure, total norepinephrine dose, and total cannulation time.

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Conclusions: New-onset atrial arrhythmias are a frequent complication during VV-ECMO and are independently associated with excessive in-hospital mortality. Thus, their presence may serve as an important prognostic tool in this patient population.
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Key Words: ECMO; arrhythmia; intensive care; mortality

VENO-VENOUS extracorporeal membrane oxygenation (VV-ECMO) is an effective salvage therapy for the management of severe respiratory disorders including acute respiratory distress syndrome (ARDS),[1-4] primary graft failure after lung transplantation,[5] trauma-induced lung injury,[6] and exacerbations of chronic lung diseases.[7,8] Although VV-ECMO may improve survival in refractory respiratory disease,[9] prolonged courses frequently result in deleterious hemorrhagic, thromboembolic, neurologic, and infectious complications, among others.[10-12]

Anecdotally, the authors have observed a high incidence of atrial arrhythmias in VV-ECMO patients at their center. Critically ill patients are at high risk of developing new-onset atrial arrhythmias during their hospitalization.[13-18] Previously identified risk factors include receipt of catecholamines, increased age, male sex, electrolyte abnormalities, prior cardiovascular disease, acute renal failure, and higher severity of illness on presentation.[14,19,20] Inflammation, even in the absence of infection, plays a role in the development of atrial arrhythmias via direct oxidative damage to atrial myocytes.[21] Hypoxemia, a frequent occurrence in patients requiring VV-ECMO, also contributes to atrial arrhythmias, as altered oxygen delivery can trigger cardiac remodeling and fibrosis.[22] Timely management of atrial arrhythmias is imperative, as their development previously has been associated with prolonged hospitalization, increased healthcare resource allocation, and mortality.[17,23]

To the authors' knowledge, no prior studies have investigated the incidence of new-onset atrial arrhythmias in patients treated with VV-ECMO and their association with outcomes. Given that atrial arrhythmias can be associated with hemodynamic compromise and frequently occur during periods of severe physiologic stress, the authors hypothesized that new-onset atrial arrhythmias would be independently associated with in-hospital mortality in VV-ECMO patients. Secondary objectives were to characterize the overall incidence of atrial arrhythmias, ascertain the impact of atrial arrhythmias on thrombotic risk, and identify potential risk factors for the development of atrial arrhythmias.

Methods

Study Population, Data Collection, and Definitions

The institutional review board at the University of Maryland, Baltimore approved this study. All patients 18 years and older who were treated with VV-ECMO at the University of Maryland Medical Center between January 1, 2016, and January 1, 2019, were included. Patients were identified through an institutional ECMO database, and data were extracted from electronic medical records. Data from the patient's first ECMO run were included in the final analysis, and those from subsequent runs were excluded. Patients also were excluded if they were cannulated or converted to venoarterial (VA) or veno-arteriovenous (VA-V)-ECMO during their admission or were treated with ECMO for fewer than 24 hours. Patients were stratified by whether they experienced new-onset atrial arrhythmias during the VV-ECMO run. Patients with a documented history of paroxysmal atrial fibrillation or atrial flutter were included in the analysis if they were in normal sinus rhythm before and at the time of VV-ECMO cannulation. Patients with permanent atrial fibrillation or atrial flutter, or those in an atrial arrhythmia at the time of VV-ECMO cannulation, were excluded.

Baseline demographics, comorbidities, VV-ECMO indication, hospital and intensive care unit (ICU) length of stay, laboratory values, medication administration, and clinical outcomes were collected retrospectively from ICU progress notes and flowsheets. Baseline characteristics were used to calculate Respiratory Extracorporeal Membrane Oxygenation Survival Prediction (RESP) scores for each patient, the methods of which previously have been described.[24] Lower RESP scores are associated with a higher probability of mortality in VV-ECMO patients.

Definitions

New-onset atrial arrhythmias were defined as either atrial fibrillation or atrial flutter that began after cannulation for VV-ECMO and resulted in rapid ventricular rate (heart rate >100 beats per minute) or systemic hypotension (mean arterial pressure <65 mmHg). Only atrial arrhythmias that required electrical or pharmacologic intervention were included in the analysis, as these were deemed to be clinically relevant. The incidence of atrial arrhythmia was captured from documentation in ICU progress notes, clinical event notes, or via electrocardiograms uploaded into the patient's chart. Acute ischemic stroke was defined as a new focal neurologic deficit persisting >24 hours with confirmatory computed tomography imaging. Acute renal failure was defined as the need for renal replacement therapy due to electrolyte derangement, volume overload, or refractory acidosis. On-pump arterial thrombosis was defined as any evidence of acute thrombosis during ECMO and confirmed by computed tomography, ultrasonography, or echocardiography. Abnormal left atrial diameter was determined by transthoracic echocardiography and defined as a 2-dimensional–derived anteroposterior linear left atrial diameter >4.1 cm for men and >3.9 cm for women.[25]

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Anticoagulation, Transfusion, and ECMO Details

Patients received anticoagulation with intravenous unfractionated heparin, targeting a goal-activated partial thromboplastin time of 45-to-55 seconds. Anticoagulation pauses or alteration in activated partial thromboplastin time goals were guided by ICU providers and surgeons and were based on patients' clinical status and bleeding risk. Argatroban or bivalirudin replaced unfractionated heparin in the setting of suspected or confirmed heparin-induced thrombocytopenia, or in cases of refractory heparin resistance. Blood transfusions were at the discretion of ICU providers. In most patients, a hemoglobin $\geq$7 g/dL was targeted. Platelet counts were maintained $\geq$40,000 K/μL. VV-ECMO flow was titrated to a goal arterial partial pressure of oxygen $\geq$55 mmHg and oxygen saturation $\geq$88%. Sweep was titrated to a target pH of 7.35-to-7.45 and arterial partial pressure of carbon dioxide of 35-to-45 mmHg. The ECMO circuit included a Rotaflow pump (Getinge Group, Wayne, NJ) and a Quadrox-i oxygenator (Getinge Group, Wayne, NJ). The authors' standard cannulation strategy involved a femoral venous drainage cannula with the return cannula placed in the right internal jugular vein.

Outcomes and Objectives

The study's primary outcome was in-hospital mortality. Secondary outcomes were a composite of thrombotic events, which included ischemic stroke and on-pump arterial thrombosis. Other secondary objectives included characterization of atrial arrhythmia incidence, risk factors, and management.

Statistical Analysis

Categorical variables were expressed as the number and percentage of patients, and continuous variables were expressed as the mean $\pm$ standard deviation or median (25th–75th percentile) depending on normality. Categorical variables were compared using the χ^2 test or Fischer exact test as appropriate. Parametric continuous data were analyzed using the Student *t* test. Nonparametric data were analyzed using the Wilcoxon rank-sum test.

After exploring the unadjusted relationship between atrial arrhythmias and in-hospital mortality, the authors created a multivariate logistic regression model to control for important confounders. RESP score (a composite score of 12 variables previously associated with VV-ECMO mortality),[24] acute renal failure, total norepinephrine dose administered (in mg/kg), and total cannulation time, were selected a priori as relevant covariates that would be included in the final model. Odds ratios (OR) for all variables in the final model were reported with 95% confidence intervals (CI). Correlation matrices and variance inflation factors were used to assess multicollinearity, and the Hosmer-Lemeshow Test was used to assess goodness-of-fit for the models. The area under the receiver operating characteristic (AUROC) curve was reported for the model as a measure of discrimination. The authors also created a multivariate model for development of an atrial arrhythmia during VV-ECMO. Covariates were selected for this model based on whether they had an unadjusted/crude relationship with the development of an atrial arrythmia. The same diagnostic tests were performed for this regression model.

Kaplan-Meier survival analysis was used to depict freedom from atrial arrhythmia during VV-ECMO. An a priori sample size calculation was not performed; however, the authors' sample size was large enough to detect an absolute difference of 17% for in-hospital mortality between patients with atrial arrythmias and those without. For all tests, a p value $<$0.05 was considered statistically significant. The Strengthening the Reporting of Observational Studies in Epidemiology statement for cohort studies was referenced during preparation of the manuscript. All statistical analyses were performed using R software (version 4.0.5; R Core Team; Vienna, Austria).

Results

Baseline Characteristics

During the study period, 247 patients were screened, of whom 219 met study inclusion criteria. Of the 28 patients excluded, 15 patients were on ECMO support for fewer than 24 hours, 5 patients were younger than 18, 3 patients were cannulated for either VA- or VA-V-ECMO during their admission, 4 patients lacked complete data, and 1 patient was transferred to another hospital post-cannulation (Fig 1). There were significant differences in baseline characteristics between those who had new-onset atrial arrhythmia during VV-ECMO and those who did not (Table 1). Briefly, those with new-onset atrial arrhythmias were older (52 $\pm$ 15 years *v* 43 $\pm$ 15 years; p $<$ 0.001), more likely to be male (73.1% *v* 52.6%;

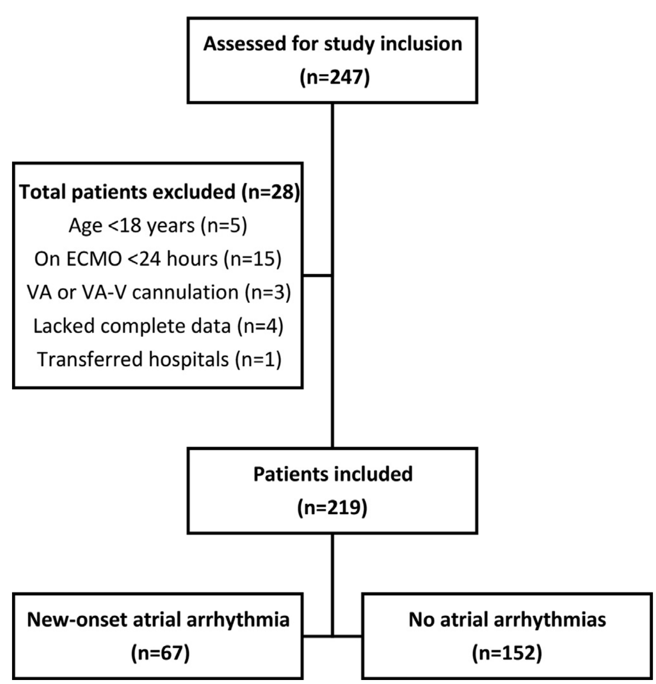

Fig 1. Flow chart of the study sample selection including reasons for patient exclusion and stratification.

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Table 1
Baseline Characteristics

Variable	Total Cohort (n = 219)	Atrial Arrhythmia (n = 67)	No Atrial Arrhythmia (n = 152)	p Value
Age (y), mean ± SD	46 ± 15	52 ± 15	43 ± 15	<0.001
Male, n (%)	129 (58.9)	49 (73.1)	80 (52.6)	0.007
Race, n (%)				0.09
White	117 (53.4)	44 (65.7)	73 (48)	
Black	80 (36.5)	17 (25.4)	63 (41.4)	
Hispanic	2 (0.9)	0 (0)	2 (1.3)	
Asian	7 (3.2)	1 (1.5)	6 (3.9)	
Other	13 (5.9)	5 (7.5)	8 (5.3)	
BMI (kg/m^2), mean ± SD	31.9 ± 9.1	32.9 ± 9.9	31.4 ± 8.8	0.27
Medical history, n (%)				
Diabetes mellitus	63 (28.8)	24 (35.8)	39 (25.7)	0.17
Hypertension	107 (48.9)	37 (55.2)	70 (46.1)	0.27
Pulmonary hypertension	16 (7.3)	9 (13.4)	7 (4.6)	0.04
Chronic lung disease	92 (42)	29 (43.3)	63 (41.4)	0.92
Chronic kidney disease	25 (11.4)	11 (16.4)	14 (9.2)	0.19
Cardiovascular disease	40 (18.3)	19 (28.4)	21 (13.8)	0.02
Paroxysmal atrial fibrillation/flutter	17 (7.8)	9 (13.4)	8 (5.3)	0.05
Cardiac surgery	24 (11)	12 (17.9)	12 (7.9)	0.05
Organ failure, n (%)				
Acute kidney injury	121 (55.3)	38 (56.7)	83 (54.6)	0.89
Renal replacement therapy	56 (25.6)	20 (29.9)	36 (23.7)	0.43
LVEF ≤40%	37 (16.9)	15 (22.4)	22 (14.5)	0.23
RV dysfunction, n (%)				0.54
None	130 (59.4)	37 (55.2)	93 (61.2)	
Mild	30 (13.7)	12 (17.9)	18 (11.8)	
Moderate	22 (10)	7 (10.4)	15 (9.9)	
Severe	12 (5.5)	5 (7.5)	7 (4.6)	
Smoking history, n (%)				0.32
Never	95 (43.4)	25 (37.3)	70 (46.1)	
Former	63 (28.8)	23 (34.3)	40 (26.3)	
Active	54 (24.7)	14 (20.9)	40 (26.3)	
Alcohol history, n (%)				0.41
Never	143 (65.3)	39 (58.2)	104 (68.4)	
Former	6 (2.7)	3 (4.5)	3 (2)	
Active	62 (28.3)	20 (29.9)	42 (27.6)	
Cause of respiratory failure, n (%)				0.93
ARDS secondary to infection	118 (53.9)	36 (53.7)	82 (53.9)	
Primary graft dysfunction	16 (7.3)	6 (9)	10 (9.2)	
Pulmonary embolism	5 (2.3)	2 (3)	3 (2)	
Bridge to transplant	13 (5.9)	4 (6)	9 (5.9)	
ARDS secondary to noninfectious cause	22 (10)	8 (11.9)	14 (9.2)	
Trauma	13 (5.9)	3 (4.5)	10 (6.6)	
Exacerbation of chronic lung disease	8 (3.7)	1 (1.5)	7 (4.6)	
Other	24 (11)	7 (10.4)	17 (11.2)	
Left atrial dimension abnormal, n (%)	32 (14.6)	13 (19.4)	19 (12.5)	0.32
Post-cardiotomy, n (%)	15 (6.8)	6 (9)	9 (5.9)	0.40
RESP score at cannulation, mean ± SD	2.1 ± 4.1	1 ± 3.3	2.5 ± 4.3	0.007
Medications prior, n (%)				
Beta-blocker	31 (14.2)	16 (23.9)	15 (9.9)	0.01
Calcium channel blocker	17 (7.8)	9 (13.4)	8 (5.3)	0.05
Digoxin	5 (2.3)	2 (3)	3 (2)	0.64
Antiarrhythmic	2 (0.9)	0 (0)	2 (1.3)	0.99

Abbreviations: ARDS, acute respiratory distress syndrome; BMI, body mass index; LVEF, left ventricular ejection fraction; RESP, Respiratory Extracorporeal Membrane Oxygenation Survival Prediction; RV, right ventricle.

p = 0.007), and had lower RESP scores (1 ± 3.3 *v* 2.5 ± 4.3; p = 0.007). Additionally, those with new-onset atrial arrhythmias were more likely to have a prior history of pulmonary arterial hypertension (13.4% *v* 4.6%; p = 0.04) and cardiovascular disease (28.4% *v* 13.8%; p = 0.02). History of paroxysmal atrial arrhythmias and cardiac surgery also were more frequent in the new-onset atrial arrhythmia group, although these differences were not statistically significant. The majority of patients were cannulated for VV-ECMO due to refractory ARDS secondary to infection. Significantly more patients were taking a beta-blocker or calcium channel blocker before admission in the atrial arrhythmia group.

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Table 2
Atrial Arrhythmia Characteristics

Variable	Atrial Arrhythmia (n = 67)
Time to first atrial arrhythmia (d), median (IQR)	1.7 (0.8-5.4)
Duration of atrial arrhythmia (d), median (IQR)	1 (1-2)
Treatment, n (%)	
Beta-blocker	35 (52.2)
Calcium channel blocker	9 (13.4)
Digoxin	16 (23.9)
Amiodarone	57 (85.1)
Direct current cardioversion	39 (58.2)
Total amiodarone dose (mg), median (IQR)	2207.5 (864.4-4771)
Number of cardioversions, median (IQR)	2 (1-4.5)
Time to treatment (d), median (IQR)	
Beta-blocker	0.6 (0-2.7)
Calcium channel blocker	0.9 (0.5-3.3)
Digoxin	2.3 (0.5-6.3)
Amiodarone	0.3 (0-0.6)
Electrolytes before atrial arrhythmia, mean ± SD	
Potassium (mmol/L)	4.2 ± 0.6
Magnesium (mg/dL)	2.4 ± 0.3

Abbreviations: IQR, interquartile range; SD, standard deviation.

Atrial Arrhythmia Characteristics

A total of 67 patients (30.5%) experienced new-onset atrial arrhythmias post−ECMO cannulation. Table 2 details the atrial arrhythmia characteristics and management. Figure 2 depicts the freedom from atrial arrhythmia during VV-ECMO. The time to first atrial arrhythmia was 1.7 (interquartile range [IQR], 0.8-5.4) days, and arrhythmias lasted a median of 1 (IQR, 1-2) day. Amiodarone was the most frequently used medication for atrial arrhythmia management (85.1%), followed by beta-blockade (52.2%). The median total amiodarone dose administered was 2.2 (IQR, 0.9-4.7) grams. More than half of patients (58.2%) required direct-current cardioversion to resolve the episode, and many patients required multiple cardioversions (median 2 [IQR, 1-4.5]). The mean serum potassium and magnesium before the first atrial arrhythmia episode were 4.2 ± 0.6 mmol/L and 2.4 ± 0.3 mg/dL, respectively.

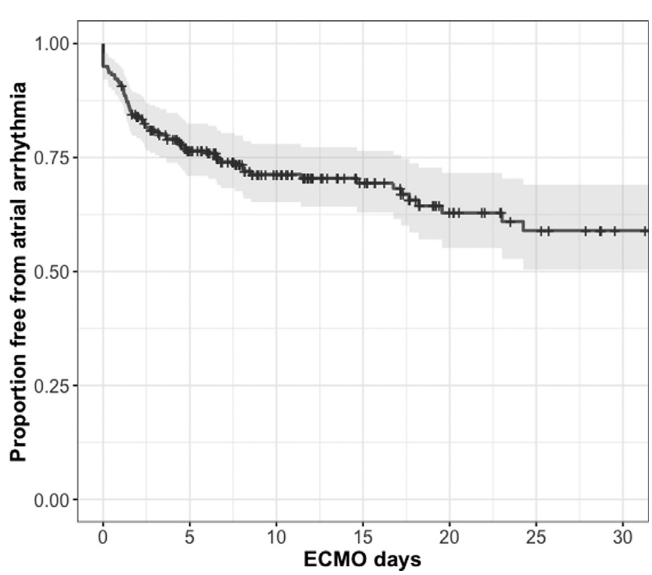

Fig 2. Kaplan-Meier survival curve depicting proportion of patients free from new-onset atrial arrhythmias.

ECMO Characteristics and Outcomes

Study outcomes are detailed in Table 3. Patients who experienced new-onset atrial arrhythmias had longer ICU length of stay (40.5 *v* 30.3 days; p = 0.05) and ECMO duration (14.9 *v* 10.9 days; p = 0.05). Significantly more patients who experienced new-onset arrhythmia received norepinephrine than those who did not (97.3% *v* 86.5%; p = 0.01). Unadjusted in-hospital mortality was significantly higher in the atrial arrhythmia group (38.8% *v* 19.1%; p = 0.003). In the multivariate logistic regression analysis, atrial arrhythmias during VV-ECMO were independently associated with an increased odds of in-hospital mortality (OR, 2.11; 95% CI, 1.02-4.37; p = 0.04), after controlling for RESP score, acute renal failure, total norepinephrine dose administered, and total cannulation time. The AUROC for this model was 0.81. The full multivariate analysis for in-hospital mortality is detailed in Table 4. There were no significant differences in the incidence of acute ischemic stroke or arterial thrombosis between the groups.

Risk Factors for Atrial Arrhythmias

Age, male sex, RESP score, home beta-blocker use, prior cardiovascular disease, prior cardiac surgery, prior atrial arrhythmia, prior pulmonary hypertension, and norepinephrine during ECMO were associated with new-onset atrial arrhythmia in the univariate analysis (Table 5). After inputting these 9 covariates into a multivariate logistic regression model, only age, male sex, and norepinephrine use were independently associated with atrial arrhythmias. The AUROC for this model was 0.77.

Discussion

In a single-center cohort of 219 patients on VV-ECMO, new-onset atrial arrhythmias were independently associated with in-hospital mortality, even after controlling for illness severity and other previously identified risk factors for mortality.[24,26] Although atrial arrhythmias have been shown previously to be associated with mortality in critical illness,[14,15,17,18] to the authors' knowledge this was the first study to demonstrate their independent association with mortality in VV-ECMO.

Atrial arrhythmias are common in the ICU population, serving as both a marker of illness severity and a predictor of poor prognosis.[13-16,18] Despite the authors' sample composed of patients with primary respiratory failure, 31% experienced new-onset atrial arrhythmias, consistent with the incidence in other critically ill populations.[18] No prior literature has identified atrial arrhythmias as a significant complication during VV-ECMO; thus, these findings revealed an underappreciated obstacle in the care of this critically ill patient population.

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Table 3
ECMO characteristics and outcomes

Variable	TotalCohort (n = 219)	Atrial Arrhythmia (n = 67)	No Atrial Arrhythmia (n = 152)	p Value
ICU length of stay (d), median (IQR)	24.7 (14.7-46.9)	26.8 (16.2-52.3)	23 (14-41.7)	0.10
Hospital length of stay (d), median (IQR)	31.6 (18.3-56)	40.5 (19.9-65.7)	30.3 (17.4-53.7)	0.05
Cannulation time (d), median (IQR)	11.6 (6.7-20.4)	14.9 (8.8-30.9)	10.9 (6.1-20.1)	0.05
Vasoactive drugs, n (%)				
Norepinephrine	196 (89.5)	65 (97)	131 (86.2)	0.02
Epinephrine	135 (61.6)	44 (65.7)	91 (59.9)	0.51
Dobutamine	23 (10.5)	5 (7.5)	18 (11.8)	0.47
Milrinone	10 (4.6)	3 (4.5)	7 (4.6)	0.99
Acute ischemic stroke, n (%)	4 (1.8)	2 (3)	2 (1.3)	0.59
Arterial thrombosis, n (%)	18 (8.2)	6 (9)	12 (7.9)	0.79
Mortality, n (%)	55 (25.1)	26 (38.8)	29 (19.1)	0.003

Abbreviations: ECMO, extracorporeal membrane oxygenation; ICU, intensive care unit; IQR, interquartile range.

Table 4
Multivariate Logistic Regression Model for In-hospital Mortality

Multivariate Model for In-hospital Mortality AUROC = 0.81 Variable	Odds Ratio With 95% CI	p Value
Atrial arrhythmia during ECMO	2.11 (1.02-4.37)	0.04
RESP score (per 1 point increase)	0.84 (0.75-0.92)	<0.001
Acute renal failure	4.20 (2.02-8.88)	<0.001
Total cannulation time (per 1 d increase)	1.00 (0.99-1.01)	0.28
Total norepinephrine dose (per 1 mg/kg increase)	1.01 (1.00-1.01)	0.02

Abbreviations: AUROC, area under the receiver operating characteristic; CI, confidence interval; ECMO, extracorporeal membrane oxygenation; RESP, Respiratory Extracorporeal Membrane Oxygenation Survival Prediction.

Receipt of catecholamines, increased age, and male sex previously have been identified as risk factors for atrial arrhythmias in critical illness.[14,19] This study confirmed their independent association with atrial arrhythmias in a VV-ECMO population. Although male sex and age are nonmodifiable risk factors, norepinephrine exposure potentially can be minimized with vasoactive agents that lack β-receptor affinity (eg, vasopressin). Epinephrine, dobutamine, and milrinone were not associated with arrhythmias in this study; however, few patients required inotropes, and the authors did not assess the impact of vasoactive dose and duration on arrhythmia occurrence. It is also possible that patients who received norepinephrine had more severe hypotension or shock, and this could have confounded the relationship between norepinephrine use and atrial arrhythmias. Home beta-blocker therapy, RESP score, history of cardiovascular disease, history of cardiac surgery, history of atrial arrhythmias, and history of pulmonary hypertension were identified as potential risk factors; however, their independent association with atrial arrhythmias diminished once multivariate modeling was performed. Analyses of larger cohorts are warranted to confirm these findings and help clinicians risk-stratify patients at time of ECMO cannulation.

The high incidence of atrial arrhythmias found in the authors' cohort may have alternative pathophysiologic explanations. Cardiac structural and electrical remodeling may occur due to infection and inflammation, processes that are prevalent during critical illness and possibly exacerbated by the proinflammatory state induced by VV-ECMO.[13,27] Structural changes, possibly related to high blood flow rates from large-diameter return cannulae could potentially induce structural changes, leading to formation of a pathophysiologic arrythmogenic substrate in atrial tissue. Atrial stretch, potentially induced by right heart dysfunction, also might lead to atrial structural remodeling and arrythmogenic atria. Hypoxemia may lead to myocyte hypoxia in atrial tissue, altering ion

Table 5
Univariate and Multivariate Regression Model for Atrial Arrhythmia*

Variable	Univariate Odds Ratio With 95% CI	p Value	Multivariate Odds Ratio With 95% CI	p Value
Age	1.04 (1.02-1.07)	<0.001	1.04 (1.01-1.06)	0.008
Male sex	2.45 (1.33-4.68)	0.005	2.66 (1.36-5.41)	0.005
RESP score	0.91 (0.84-0.98)	0.02	0.97 (0.88-1.06)	0.45
Home beta-blocker use	2.87 (1.32-6.27)	0.008	2.02 (0.80-5.11)	0.13
History of cardiovascular disease	2.47 (1.22-5.00)	0.01	0.84 (0.31-2.20)	0.73
History of cardiac surgery	2.55 (1.07-6.07)	0.03	1.20 (0.37-3.79)	0.76
History of atrial arrhythmia	2.79 (1.02-7.78)	0.04	1.29 (0.39-4.24)	0.67
History of pulmonary arterial hypertension	3.21 (1.15-9.38)	0.03	2.81 (0.82-10.57)	0.11
Norepinephrine during cannulation	5.21 (1.47-33.20)	0.03	8.96 (2.15-65.57)	0.009

Abbreviations: CI, confidence interval; RESP, Respiratory Extracorporeal Membrane Oxygenation Survival Prediction.
*Area under the receiver operating characteristic = 0.77.

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handling and causing electrical remodeling. Finally, infectious agents, such as bacteria and viruses, may cause direct myocardial injury, inducing both structural and electrical remodeling.

In the authors' cohort, atrial arrhythmias occurred early postcannulation (median 1.7 days). During this period, patients may be in the most acute phase of their illness, with high catecholamine exposure, severe metabolic derangement, fluid overload, and excessive physiologic stress. In this study, there were no differences in serum potassium and magnesium immediately before arrhythmia onset. The authors hypothesized that hypoxemia, high catecholamine exposure, and inflammation may have contributed to arrhythmogenesis. Although the duration of atrial arrhythmias was relatively short in this study, hemodynamic depression might not improve immediately upon arrhythmia termination.[28] Also, although there was no significant difference in overt strokes, it is possible that subclinical strokes occurred in patients who had atrial arrhythmias.

In addition to excess mortality, atrial arrhythmias also may be associated with increases in healthcare expenditure and resource allocation.[17,29,30] This study demonstrated that those who experienced new-onset atrial arrhythmia had longer hospital length of stay and longer durations of ECMO. High-quality ECMO care already is resource-intensive, with mean hospital cost in excess of \$200,000.[31] Thus, mitigating the severity and incidence of atrial arrhythmias might be helpful in reducing ECMO's total cost. Additionally, the authors' results highlighted that these arrhythmias are challenging to manage. Although many of the arrhythmias resolved within 1 day, they frequently were refractory to standard pharmacologic therapies (eg, beta-blockers and amiodarone). More than half of the authors' cohort required at least 1 electrical cardioversion, which has its own associated risks.

This study had several limitations that are worth highlighting. First, this study was conducted at a single, high-volume ECMO center. ECMO care, circuit configuration, ICU management, and patient populations may vary among centers, thus limiting the generalizability of the authors' findings. Additionally, there was no standard protocol for treatment of atrial arrhythmias. Treatment depended on individual provider practice, and, thus, the authors cannot comment on the effectiveness of the treatment administered. Second, this study was retrospective and prone to confounding. The authors attempted to control for relevant covariates associated with excess in-hospital mortality in VV-ECMO; however, residual confounding is possible, particularly severity of patients' initial shock and illness course during ECMO. It is possible that atrial arrhythmias served as an early marker of illness severity, and, thus, may have been helpful in prognostication on top of previously validated scoring tools.[24] The authors' model's AUROC of 0.81 suggested good discrimination for mortality, but the authors did not validate their findings in an external cohort. Third, the authors excluded 15 patients with <24 hours of VV-ECMO support. This was done due to lack of complete documentation for clinical outcomes and laboratory values; however, this may have biased their findings. Fourth, the authors did not have access to all potential risk factors for atrial arrhythmias, such as thyroid disease and pulmonary embolism, as consistent data were lacking. Fifth, abnormal left atrial diameter was measured by transthoracic echocardiography and defined as a 2-dimensional−derived anteroposterior linear left atrial diameter >4.1 cm for men and >3.9 cm for women. Current guidelines recommend measuring left atrial volume using a biplane disk summation technique (34 mL/m^2 is considered abnormal for both sexes), as this is more prognostic of cardiovascular outcomes.[25] Left atrial volume was not detailed in echocardiography reports at the authors' institution during the study period. Thus, it is possible that left atrial dilation may have contributed to new-onset atrial arrhythmias. Lastly, the authors' definition of atrial arrhythmias only included those that resulted in hemodynamic derangement, requiring electrical or pharmacologic intervention. This likely led to an underrepresentation of the true disease burden. Due to the retrospective nature of this study, the incidence and duration of these events had to be extracted from progress notes, event notes, and uploaded electrocardiograms. It is likely that some arrhythmias were not reported in the medical record. It is unclear if subclinical atrial arrhythmias are independently associated with in-hospital mortality.

Conclusion

In this cohort of patients with respiratory failure requiring VV-ECMO support, new-onset atrial arrhythmias were a common complication during VV-ECMO and were independently associated with excess in-hospital mortality and a prolonged ECMO course. Thus, the occurrence of new-onset atrial arrythmias may have prognostic value in this population. Whether atrial arrhythmias during VV-ECMO may be modifiable with prophylaxis in high-risk patients (eg, patients requiring high levels of vasopressor support, older patients, etc) is of interest. Multicenter studies are needed to better understand the incidence of atrial arrhythmias during VV-ECMO and how it impacts both short-term and long-term patient outcomes.

Conflict of Interest

None.

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